

軟體測試與資安檢測

教師：王凡

課程代號：EE 5191

授課時間：每週三 9：10—12：00

學分：3

博理館 112

民國 112 年秋季班

期末考

考試中，請勿參閱資料。

In **Figure 1** is a pseudo-code for quicksort. There are two procedures in this code.

```
// Sorts a (portion of an) array, divides it into partitions, then sorts those
algorithm quicksort(A, lo, hi) is
  // Ensure indices are in correct order
  if lo >= hi || lo < 0 then
    return
  // Partition array and get the pivot index
  p := partition(A, lo, hi)
  // Sort the two partitions
  quicksort(A, lo, p - 1) // Left side of pivot
  quicksort(A, p + 1, hi) // Right side of pivot

// Divides array into two partitions
algorithm partition(A, lo, hi) is
  pivot := A[hi] // Choose the last element as the pivot
  i := lo - 1 // Temporary pivot index
  for j := lo to hi - 1 do
    // If the current element is less than or equal to the pivot
    if A[j] <= pivot then
      // Move the temporary pivot index forward
      i := i + 1
      // Swap the current element with the element at the temporary pivot index
      swap A[i] with A[j]

  // Move the pivot to the correct position (between the smaller and larger elements)
  i := i + 1
  swap A[i] with A[hi]
  return i // the pivot index
```

Figure 1: a pseudo-code for quicksort

1. Draw the control-flow graph of the above program. There are various ways to draw graph elements with respect to (recursive) procedure calls. Please argue why you draw it in the way of your answer below. Note that the answers to the subsequent questions rely on the control flow graph you draw below. (5pts/5)

2. List the test requirements for prime-path coverage (PPC) continuing from your answer to question 1. (5pts/10)

3. Continuing from question 2, please write down a test set (i.e, input values of the list A, parameters lo and hi) for the full PPC. (5pts/15)

4. Continuing from question 2, please note that variables lo and hi are formal parameters which are instantiated to actual parameters for each call to the child procedures. Please show your creativity by defining and writing down a dataflow criterion for managing instances of the actual parameters for a strong enough and not too expensive test plan. Please also argue for your definition. (5pts/20)

5. Continuing from question 4, please write down a set of du-paths that satisfies all-defs coverage with respect to parameter instances of `lo` and `hi` according to your data-flow criterion. (Direct tours only.) (5pts/25)

6. Continuing from question 4, please write down a minimal test set (i.e, input values of list A and parameters lo and hi) that satisfies all-uses coverage with respect to parameter instances of lo and hi according to your dataflow test criterion. (Direct tours only.) (5pts/30)

7. Consider the following predicate

$$(a < 2 \parallel !r) \ \&\& \ (a > b \parallel s) \ \&\& \ (b \neq 100 \parallel r \parallel s).$$

Please compute (and simplify) the conditions under which each of the atoms (clauses in textbook terms) that determines this predicate. (5pts/35)

8. Continuing from question 7, write down the complete truth table for all clauses (atoms in standard logic terms). (5pts/40)

9. Continuing from question 8, please identify all pairs of rows from your truth table that satisfy restricted active clause coverage (RACC) with respect to each clause (atoms in standard logic terms). (5pts/45)

10. Continuing from question 8, identify all rows from your table that satisfy *implicant coverage (IC)* with respect to each implicant. (5pts/50)

11. Assume that we have an app login panel with the following screen layout.

The image shows a login panel with a blue background and rounded corners. It contains the following elements:

- Account (email):** A text label followed by a white rectangular input field with a red border.
- Password:** A text label followed by a white rectangular input field with a red border.
- Captcha:** A text label followed by a white rectangular input field with a red border.
- License Agreement:** A checkbox with a red border, followed by the text "Check to agree all the license terms!".
- Captcha Image:** A square image showing a distorted captcha with the characters "V6TqJBCDS".
- Buttons:** Three oval buttons with red borders and light gray backgrounds, stacked vertically on the right side:
 - Top button: "X" (close button).
 - Middle button: "login".
 - Bottom button: "New captcha".
- Forgot Password:** An oval button with a red border and light gray background, located below the "New captcha" button, containing the text "Forget password".

There is no other document. Please define characteristics of inputs to the web interface and list them. You must write down a proof for the completeness and disjointness of the partitions by your characteristics. (5pts/55)

12. Continuing from question 11, please write down three input values (including at least one boundary value) for each block of each characteristic. (5pts/60)

13. Continuing from question 12, please write down your pairwise test suite. (5pts/65)

14. Continuing from question 11, please write down three SQL-injection test input to the login window and explain why they may break in and cause information leakage.
(5pts/70)

15. What are the five levels of test process maturity? (5pts/75)

16. Assume that we have the procedure in Figure 1 to test. Please write down a mutant with the Arithmetic Operator Replacement (AOR) and write down a test case (input list A , lowbound lo , and upperbound hi) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/80)

17. In Sangaroonsilp, Dam, and Ghose's 2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE) paper "*On Privacy Weaknesses and Vulnerabilities in Software Systems*," what is the authors' argument that CVE and CWE are not good enough for classifying privacy vulnerabilities? (5pts/85)

18. The 2022 ACM SIGSAC Conference on Computer and Communications Security paper: “*StrongBox: A GPU TEE on Arm Endpoints*” by Deng, Wang, Yu, et al was built around the technology of ARM TrustZone and TZASC. Please explain ARM TrustZone and TZASC. (5pts/90)

19. According to Kappos, Yousaf, Stütz, Rollet, Haslhofer, and Meiklejohn's paper in 31st USENIX Security Symposium: "*How to peel a Million: Validating and Expanding Bitcoin Clusters*," what are the techniques to make blockchain and cryptocurrency less anonymous? (5pts/95)

20. In the ICSE 2023 paper: “*Heterogeneous Anomaly Detection for Software Systems via Semi-supervised Cross-modal Attention*” by Lee, Yang, Chen, Su, Yang and Lyu, the authors proposed to use logs and metrics for better detection of system anomalies. Please explain what metrics they referred to and how the metrics can be used for their purpose. (5pts/100)